

Consumer Lending Risk Insights Through Data-Driven Analytics

1. Background & Business Context

Consumer finance companies face significant challenges when lending money to individuals with **limited, insufficient, or non-existent credit history**. A lack of credit information increases the uncertainty involved in predicting whether a customer will repay the loan on time.

Loan defaults lead to direct financial losses and increase the company's risk exposure. Conversely, incorrectly rejecting a potentially good customer results in lost business opportunities. Therefore, data-driven decision-making is essential for:

- **Reducing credit risk**
- **Improving loan approval strategies**
- **Identifying high-risk applicants early**
- **Setting appropriate credit limits and interest rates**

Your role is to perform **Exploratory Data Analysis (EDA)** and **Statistical Hypothesis Testing** to uncover the **drivers of loan default** using real-world customer data.

The assignment simulates work typically done by analysts in risk management teams at banks, NBFCs, and fintech lenders.

2. Objective of the Case Study

The main aim is to:

- Understand the customer attributes and loan characteristics
- Identify behavioral and demographic variables that drive default
- Discover patterns that differentiate “defaulters” from “non-defaulters”
- Use hypothesis testing to validate whether observed patterns are statistically significant

- Provide actionable recommendations that reduce credit risk

You must work with two provided datasets that capture customer information at the time of applying for the loan and their previous loan histories.

3. Dataset Description

You are provided with **three files**:

3.1 Dataset 1 — **credit_risk_applicants.csv**

This dataset includes information about **current loan applicants** and contains:

a. Target variable

- **TARGET = 1** → The client had payment difficulties (defaulted or significantly delayed installment payments)
- **TARGET = 0** → The client repaid on time

b. Demographics & Personal Information

- **CODE_GENDER** (Gender)
- **DAYS_BIRTH** (Age in days; negative values)
- **NAME_EDUCATION_TYPE**
- **NAME_FAMILY_STATUS**
- **CNT_CHILDREN**
- **CNT_FAM_MEMBERS**

c. Financial Variables

- **AMT_INCOME_TOTAL** (Annual Income)

- `AMT_CREDIT` (Loan amount requested)
- `AMT_ANNUITY`
- `AMT_GOODS_PRICE`

d. Employment & History

- `DAYS_EMPLOYED`
- `OCCUPATION_TYPE`
- `FLAG_OWN_CAR`, `FLAG_OWN_REALTY`

e. Risk & External Score Features

- `EXT_SOURCE_1`, `EXT_SOURCE_2`, `EXT_SOURCE_3`
These are external credit bureau or scoring system indicators.

f. Housing & Living Conditions

Variables such as:

- `APARTMENTS_AVG`, `YEARS_BUILD_AVG`, `LIVINGAREA_AVG`, etc.
provide information on the living conditions and real estate characteristics.

These variables may indicate stability or affordability.

3.2 Dataset 2 — `credit_risk_previous_loans.csv`

This dataset contains details about **credit risk and previous loan applications** made by the same customers (`SK_ID_CURR` is the linking key).

Key attributes include:

- `NAME_CONTRACT_STATUS` (Approved, Refused, Cancelled, Unused Offer)
- `AMT_CREDIT` (Credit applied previously)

- `AMT_APPLICATION`
- `AMT_DOWN_PAYMENT`
- `CNT_PAYMENT` (Installment count)
- `NAME_CASH_LOAN_PURPOSE`
- `RATE_INTEREST_PRIMARY`
- `DAYS_DECISION` (How many days before the current application the decision was made)

This dataset helps analyze patterns such as:

- Repeated rejections
 - High-risk behavior
 - History of applying for multiple loans
 - Past loan outcomes that may influence future defaults
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3.3 Dataset 3 — `credit_risk_metadata.csv`

This file explains the meaning, data type, and context of every variable.

Students are expected to use it for:

- Understanding variable significance
 - Choosing appropriate plots and transformations
 - Interpreting results accurately
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4. Business Challenge

You are hired as a Data Analyst in the **Credit Risk Department** of a lending company. Your task is to analyze applicant profiles to assist in the following major business goals:

4.1 Reduce the Risk of Approving High-Risk Customers

Identifying attributes that strongly correlate with defaults helps minimize bad loans.

4.2 Avoid Rejecting Creditworthy Customers

If good customers are mistakenly classified as risky, the business loses revenue.

4.3 Recommend Data-Driven Strategies

Based on your analytical findings, the company expects actionable recommendations on:

- Loan approval criteria
 - Customer profiling
 - Interest rate adjustments
 - Required documentation
 - Employment verification policies
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5. Analytical Tasks Expected from Students

You must perform the following structured analysis steps:

5.1 Data Cleaning

- Identify missing values
- Handle outliers
- Detect anomalies in variables (e.g., impossible values in `DAYS_EMPLOYED`)

- Decide which features to retain or discard based on business utility
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5.2 Exploratory Data Analysis (EDA)

Univariate Analysis

Study distribution patterns of:

- Income
- Credit amount
- Age
- Annuity
- Education
- Employment length

Bivariate Analysis

Compare defaulters vs non-defaulters across:

- Credit amount
- Income groups
- Housing type
- Occupation type
- External scores

Multivariate Analysis

Observe relationships among:

- Income × Credit × Default
- Age × Occupation × Default

- Housing conditions × External scores

Correlation Analysis

Identify the strongest relationships among numeric variables and interpret their significance.

5.3 Hypothesis Testing

Students must formulate and test hypotheses such as:

- **Do defaulters have significantly lower income than non-defaulters?**
- **Is the default rate different across genders?**
- **Are education level and default correlated?**
- **Do previous loan rejections predict higher current default probability?**
- **Is the company's default rate higher than the industry benchmark?**

These require statistical tests such as:

- T-tests
- Z-tests
- ANOVA
- Chi-square tests
- F-tests
- Proportion tests

Each test must include:

1. Null and alternate hypotheses
2. Choice of test and justification
3. Assumptions validation

4. Test statistic and p-value
 5. Final conclusion in plain business language
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6. Expected Deliverables

Students must submit:

A structured Jupyter Notebook / Python script

containing the full workflow and analysis.

A presentation summarizing findings

including:

- Key EDA insights
- Driver variables of default
- Results of hypothesis tests
- Business recommendations

Actionable credit policy strategy

based on discovered patterns.

7. Important Guidelines for Students

- Visualizations must be clear, labeled, and business-readable.
- Interpretations must connect analysis to real-world lending decisions.
- Avoid technical jargon; use business-friendly language.
- State assumptions clearly for every hypothesis test.
- Data cleaning decisions must be justified.

- Focus on insights, not just plots.
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8. Final Outcome

By the end of the project, students must be able to:

- Identify **risk indicators**
- Describe **customer behavioral patterns**
- Evaluate **loan eligibility patterns**
- Provide **data-driven lending recommendations**
- Understand how **statistical significance** validates insights
- Build an analytics mindset aligned with credit risk management